Genus-3 IC-2LP after the cubic GHS transfer: is the departure from $q^{4/3}$ an artifact of the factor base or the linear algebra?

research/ic_improve_20261014/04b-genus3-2lp-rebalance · 2026-10-15 · Conductor T-1 · run-20261015T0200Z
Protocol frozen at commit 9752513ea (sha256 633e69bb…); κ choice frozen at 288e74293

Rebuilt baseline. The study this follows up (`04-genus3-2lp`, its protocol, `RULES.md`, `examples/ic_improve_04_*`) is not in the repository (`problems/ic-improve-04-genus3-2lp-study-missing.md`). At the requester's choice the IC-2LP baseline was rebuilt; every arm here is paired with **that** baseline on the same instances, seeds and unit, not with the original's records.

1. Answer

On this rebuild the departure from 4/3 is **not** an artifact of either choice.

- **Baseline (BL, κ = 2, tree-cycle elimination + Lanczos):** total cost fits $q^{1.383}$ over 2^{14}–2^{20} (99% [1.382, 1.384]); local exponents trend +0.003 per $\log_2 q$ (99% [0.002, 0.004]). The relation phase fits $q^{1.332}$ (GTTD's 4/3); the linear algebra fits $q^{1.447}$, driven by the Lanczos row weight growing 7.8 ⟶ 14.4 from 2^{14} to 2^{20}.
- **A1, factor base retuned per size** (κ ∈ {0.5, 0.7, 1, 1.4, 2} on training curves 5–7): verdict **reduces the exponent**, E = 1.373 [1.372, 1.374], 0.010 below BL and still 0.039 above 4/3. Tuning picked κ = 1.4 at 2^{17}–2^{20}; paired saving 0.8% ⟶ 3.9%.
- **A2, structured Gaussian elimination before Lanczos** (κ = 2, BL's own relations): verdict **raises the exponent**, E = 1.419 [1.417, 1.421]; cheaper at every size (ratio 0.78 at 2^{14}, 0.91 at 2^{20}) but the saving shrinks as BL's weight catches up with A2's bounded ≈ 15–16.
- A3 (secondary, κ tuned on A2's pipeline) chose κ = 2 everywhere and is A2.

All 360 logarithms (180 test solves × 2 linear-algebra routes) satisfy $[\log]P = Q$ and equal the planted value; 210 tuning solves and 20 rho runs also verified; no job failed (max 119 MB, 490 s).

2. Unit, instances, arms

Unit: multiplications in $\mathbb{F}_q$ plus multiplications mod ℓ over the whole method (relation setup, walk, decomposition; elimination; Lanczos; verification); $S = \text{total}/\sqrt{\ell}$. Curves: hyperelliptic GHS cubic covers ($m = 3$, $r = 5$, T-2463's construction), $y^2 = f(x)$, deg 7 over $\mathbb{F}_q$, q the largest prime below 2^k, $k = 10..20$, $\#\mathrm{Jac}(C)(\mathbb{F}_q) = h\ell$ with $h \in \{4, 8\}$; 7 curves per size, seed 20261016; test curves 1–4 × targets 1–3 (12 instances per size). Rho: r-adding walk with Brent cycle detection on Jac(C), $k \le 14$, 4 walks per size.

3. Results

k	BL total	A1 (κ*)	A2 total	S_{BL}	S_{A1}	S_{A2}	S_{rho} (n=4)	BL/rho raw, renorm.
10	3.142e6	—	2.559e6	192.6	—	156.9	550.0	0.35, 0.58
11	6.765e6	—	5.323e6	207.8	—	163.5	739.1	0.28, 0.63
12	1.560e7	—	1.199e7	119.1	—	91.5	439.7	0.27, 0.35
13	3.878e7	—	2.997e7	148.0	—	114.3	582.6	0.25, 0.42
14	9.686e7	9.686e7 (2)	7.566e7	92.4	92.4	72.2	571.4	0.16, 0.25
15	2.472e8	2.472e8 (2)	1.974e8	83.4	83.4	66.6	—	—
16	6.731e8	6.731e8 (2)	5.575e8	80.3	80.3	66.5	—	—
17	1.700e9	1.687e9 (1.4)	1.436e9	101.3	100.5	85.6	—	—
18	4.387e9	4.301e9 (1.4)	3.801e9	92.4	90.6	80.1	—	—
19	1.167e10	1.136e10 (1.4)	1.036e10	86.9	84.6	77.2	—	—
20	3.049e10	2.930e10 (1.4)	2.763e10	56.8	54.6	51.5	—	—

Mean over 12 test instances (95% intervals in tables.md). S moves with ℓ across curves. Rho walks took 0.30–4.0 steps/√ℓ (means 1.6–2.8 vs ideal 1.25); "renorm." rescales rho to $\sqrt{\pi/2}$.

arm	phase	E (fit 14–20), 95%, 99%	E − 4/3	$E - E_{BL}$ (99%)	trend T (99%)
BL	total	**1.383** [1.382, 1.384] [1.382, 1.384]	+0.049	—	+0.003 [0.002, 0.004]
BL	relation	1.332 [1.331, 1.334]	−0.001	—	−0.001 [−0.003, 0.002]
BL	lin. algebra	1.447 [1.445, 1.448]	+0.113	—	+0.002 [0.000, 0.005]

A1	total	**1.373** [1.372, 1.374] [1.372, 1.374]	+0.039	−0.010 [−0.010, −0.010]	−0.001 [−0.002, 0.000]
A2	total	**1.419** [1.418, 1.420] [1.417, 1.421]	+0.086	+0.036 [+0.035, +0.037]	+0.001 [−0.001, 0.003]
A2	lin. algebra	1.599 [1.596, 1.602]	+0.266	+0.152 [+0.149, +0.155]	−0.027 [−0.032, −0.023]

k	14	15	16	17	18	19	20
A1/BL	1 (same run)	1	1	0.992	0.980	0.974	**0.961** [0.959, 0.963]
A2/BL	0.781	0.798	0.828	0.844	0.867	0.888	**0.906** [0.905, 0.907]
A2/BL lin. alg.	0.429	0.505	0.578	0.646	0.712	0.766	0.812

Paired cost ratios, geometric mean over the 12 instances (99% interval where shown; all intervals in tables.md). A2/BL at k = 10–13: 0.815, 0.787, 0.768, 0.773.

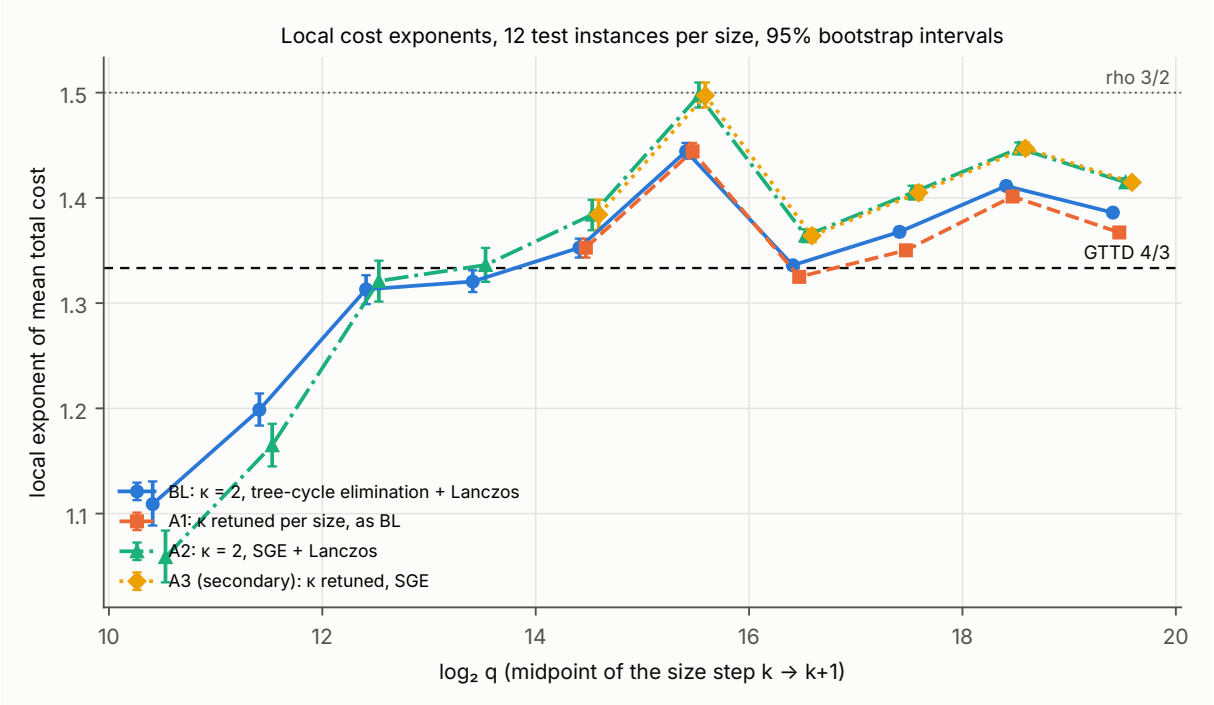


Figure 1: Local exponents of mean total cost per size step, 12 test instances, 95% paired bootstrap intervals (B = 10000). A3 coincides with A2 ($\kappa^{**} = 2$). run-20261015T0200Z.

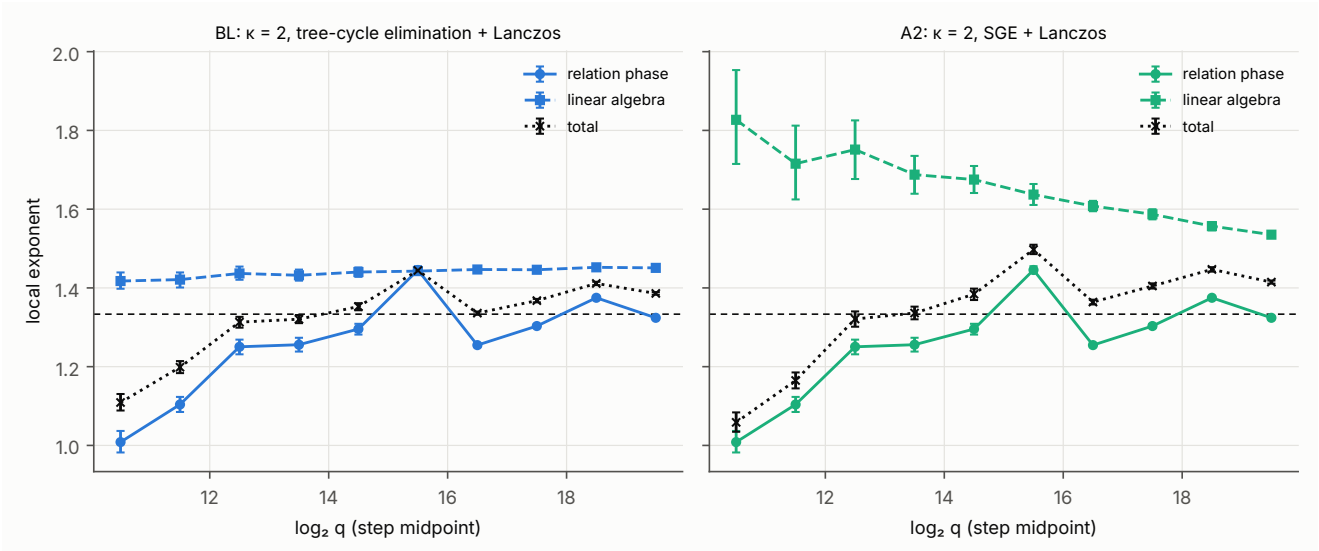


Figure 2: Local exponents by phase: the relation phase sits near 4/3 from 2^{16}; BL's linear algebra runs at ≈ 1.45; A2's linear algebra starts steeper (it removes a falling fraction of the matrix). The 15⟶16 bump is a per-trial decomposition-cost step (300 ⟶ 378 mults).

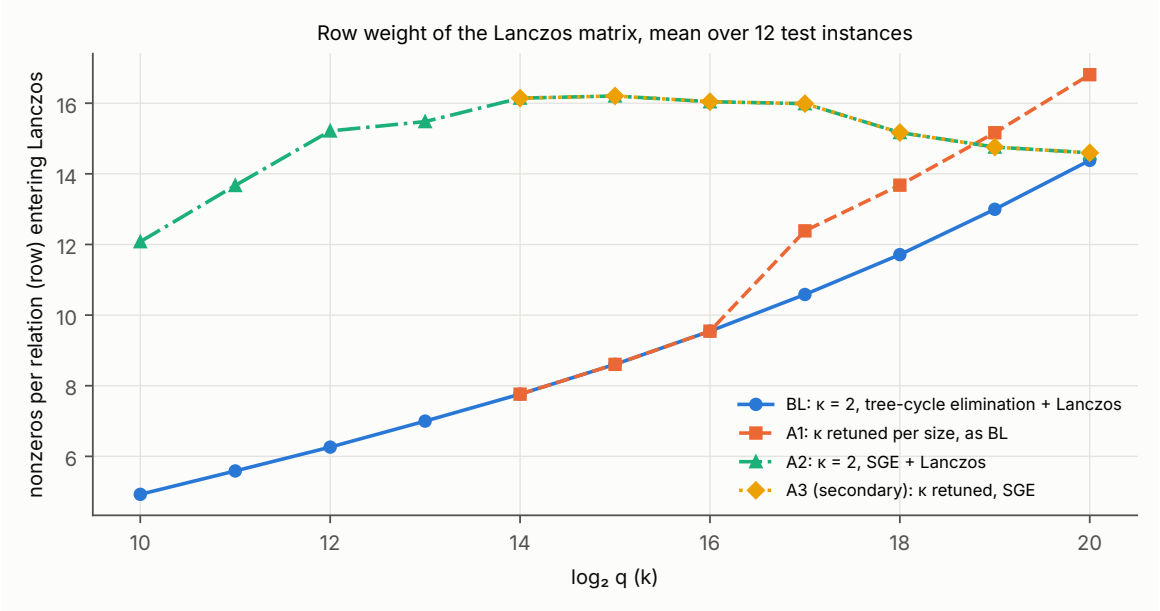


Figure 3: Nonzeros per row of the matrix entering Lanczos. BL grows 4.9 ⟶ 14.4; A2 stays 14.6–16.2 from 2^{14}; A1 at $\kappa = 1.4$ is heavier (interpretation: longer cycles in a sparser large-prime graph).

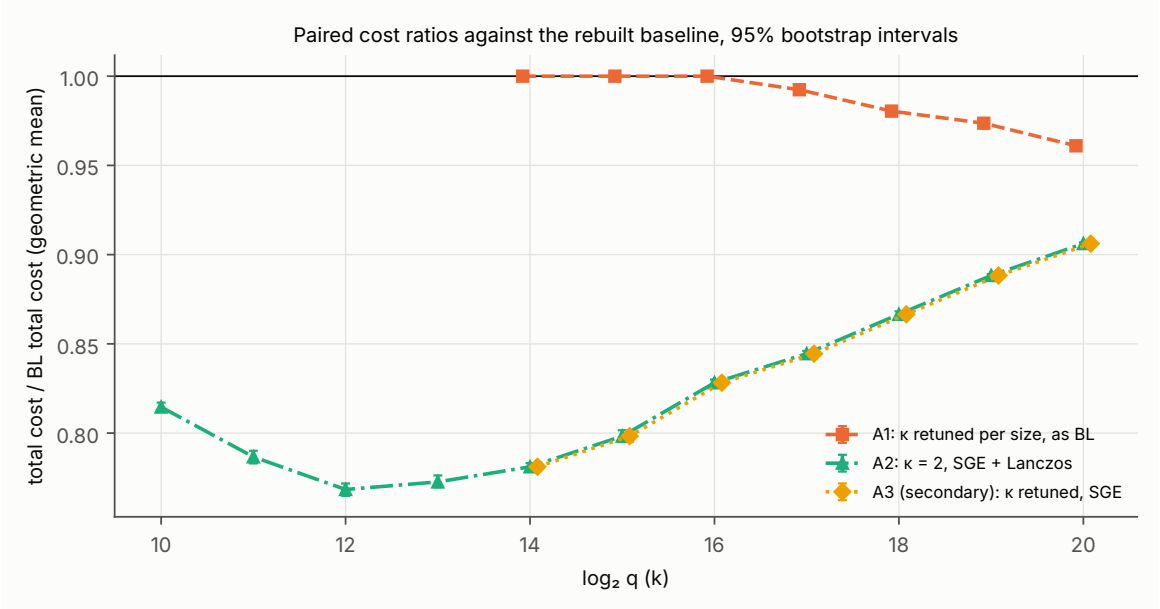


Figure 4: Paired cost ratios against BL with 95% intervals (12 instances per size).

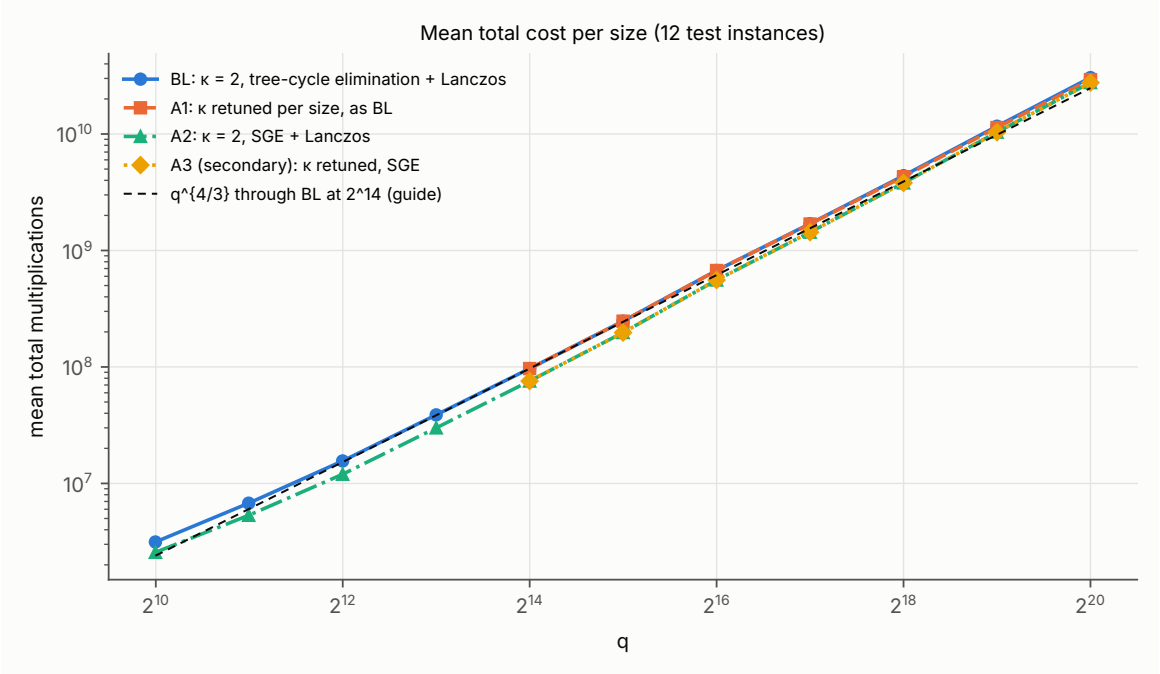


Figure 5: Mean total multiplications against q with a $q^{4/3}$ guide through BL at 2^{14} (guide only, not a fit).

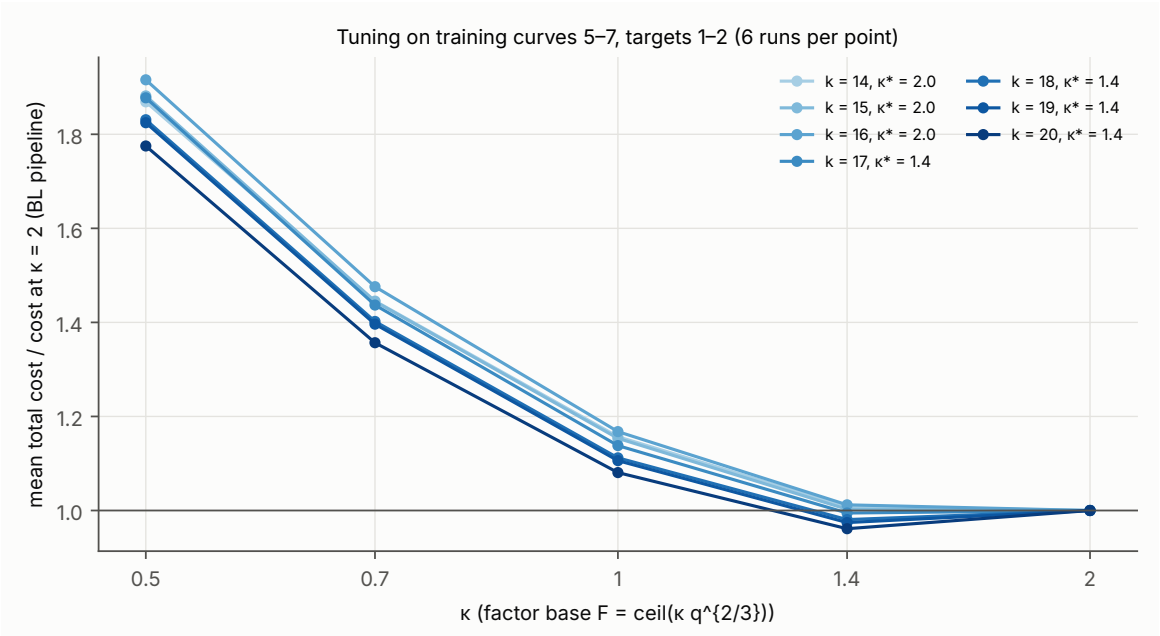


Figure 6: Tuning on training curves 5–7 × targets 1–2: mean total of the BL pipeline relative to κ = 2. The interior optimum κ = 1.4 appears from 2^{17}; at 2^{14}–2^{16} the optimum is the grid edge κ = 2.

4. Decision (PROTOCOL §7) and controls

D0 baseline departs: **yes** (99% E_{BL} [1.382, 1.384] > 4/3; 99% T_{BL} [0.0016, 0.0042] > 0). A1: **reduces the exponent** (not removes: E's 99% interval excludes 4/3). A2: **raises the exponent**. Answer: **not** an artifact of either choice on this rebuild. Power: 95% half-width of E_{BL} 0.0009 (≤ 0.03).

Controls: A2′s relation phase is BL's own record on all 132 instances (pairing). A/A: E_{BL} on target 1 minus targets 2–3 = +0.002, 95% [0.001, 0.004]. This excludes 0 but is 25× smaller than the departure, and every verdict's margin is ≥ 0.010; read as the bootstrap understating between-target spread at the 0.002 level with 4 curves per size. Lanczos accounting equals its closed form on all 360 runs. One operator error (an empty job list, not a test job) is logged in results/orchestration_errors.jsonl.

5. Interpretation (labelled)

Lanczos costs ≈ $F^2(2w+7)$; with $F \propto q^{2/3}$ and $w \propto q^{\delta}$, measured $\delta \approx 0.15$ over 2^{14}–2^{20} (tree-cycle relations lengthen as the BFS tree deepens), so the linear algebra runs above $q^{4/3}$; the constant term dilutes it to the measured 1.447. A heuristic balance with relation cost $\propto q^2/F$ gives optimal $F \propto q^{2/3}w^{-1/3}$ and total $q^{4/3}w^{1/3}$, i.e. ≈ 4/3 + δ/3 ≈ 1.38 even with continuous retuning; A1 measures 1.373 (model untested beyond the grid). A2′s Markowitz merges stop near the weight BL reaches by

2^{20}, so its saving is a constant that shrinks with q; extrapolating the ratio trend (+0.02 per size, not measured) puts parity a few sizes past 2^{20}. Only one SGE design was tested. Classification (AGENTS.md §3): A1 and A2 are engineering; neither is an advance.

Against the request's description of the original (qualitative, unpaired): local exponents 1.25 ⟶ 1.46 (original) vs 1.11 ⟶ 1.39 here; linear algebra $q^{1.52}$ vs $q^{1.447}$; relation $q^{1.30}$ vs $q^{1.332}$; row weight 7 ⟶ 25 vs 4.9 ⟶ 14.4. Same direction, smaller magnitude.

6. Limits

Õ hides log factors (decomposition costs ≈ log q per trial). The κ grid is the request's; the optimum sits on its edge κ = 2 for A2 at every size and for BL at 2^{14}–2^{16}. The transfer is not executed per target (T-2463 verified $\mathrm{Jac}(C)[\ell] = \varphi\big(E_0\big(\mathbb{F}_{q^3}\big)[\ell]\big)$); curve construction and group order are not counted. Rho is measured only at $k \leq 14$, n = 4, on Jac(C). Host: shared 4-vCPU container, no isolation; wall time is not a metric. No new theorem, so no Lean obligation.